

CoreOps Week 2 Fri Module

Kubernetes Internals & High Availability Architecture

Objective: Master control plane internals, node lifecycle, upgrade patterns, HA design, static pod behaviour, and failure-handling logic.

FRIDAY: Node pools, placement logic, PDBs, affinity/anti-affinity

Critical Concepts

- Node pool architecture
- Scheduler scoring logic
- PDBs
- Affinity vs Anti-affinity
- Taints & tolerations
- Pod spreading & topology-aware scheduling



Key Outcomes

- You can design production-grade node groups
- Diagnose why pods don't schedule
- Use PDBs to control reliability during disruptions

WHAT TO PRACTICE (HANDS-ON)

Lab 1: Create affinity and anti-affinity rules

Force pods to run on specific nodes.

Lab 2: Apply PodDisruptionBudget

Drain nodes & observe behaviour.

Lab 3: Create node taints & tolerations

WHERE TO STUDY

- Node Affinity Docs
 - PDB Docs
 - SIG-Scheduling Deep Dive (YouTube)
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INFRATHRONE MINDSET NOTES

- Production upgrades fail because of version skew.
 - Static pods are the spine of your control plane; if they break, your cluster breaks.
 - Never upgrade blindly; always rehearse upgrades in a clone cluster.
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